

EXPERIMENT 7

Determination and plotting of Capacitance of isolated and concentric sphere with different inner and outer radius and dielectric constant of the dielectric mediums.

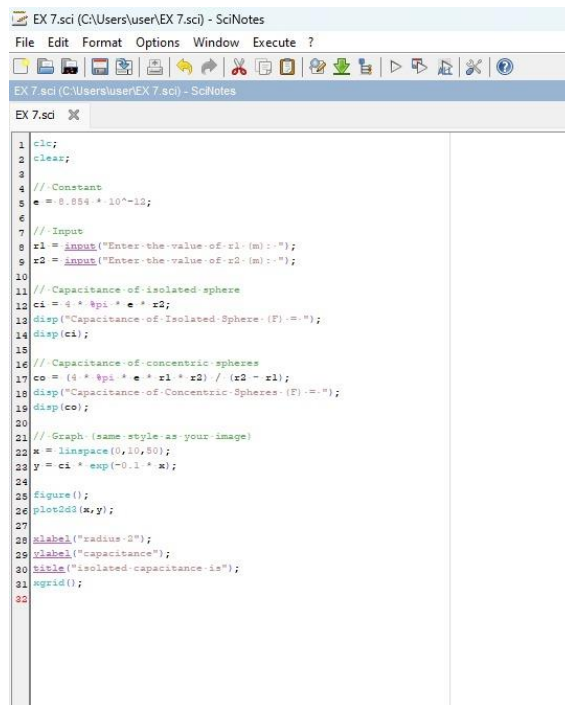
Aim:

To write program for Capacitance of an isolated sphere and two concentric sphere using SCILAB.

Software required:

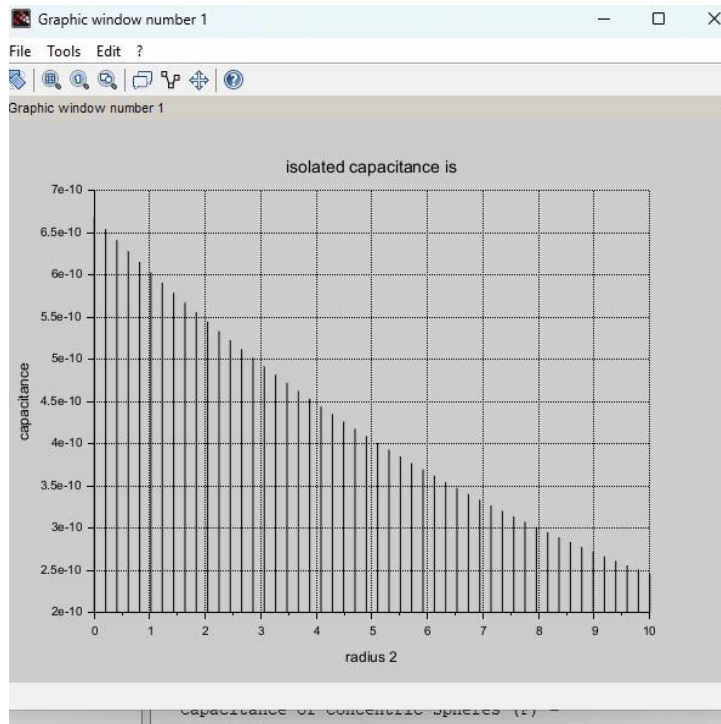
- SCILAB version 6.0.1

Program:



```
1 c1c;
2 clear;
3
4 //Constant
5 e = 8.854 * 10^-12;
6
7 // Input
8 r1 = input("Enter the value of r1. (m):-");
9 r2 = input("Enter the value of r2. (m):-");
10
11 // Capacitance of isolated sphere
12 ci = 4 * %pi * e * r2;
13 disp("Capacitance of Isolated Sphere (F) :-");
14 disp(ci);
15
16 // Capacitance of concentric spheres
17 co = (4 * %pi * e * r1 * r2) / (r2 - r1);
18 disp("Capacitance of Concentric Spheres (F) :-");
19 disp(co);
20
21 // Graph (same style as your image)
22 x = linspace(0,10,50);
23 y = ci * exp(-0.1 * x);
24
25 figure();
26 plot2d3(x,y);
27
28 xlabel("radius-2");
29 ylabel("capacitance");
30 title("isolated capacitance is");
31 xgrid();
32
```

Output:



Practical output:

```
Enter the value of r1 (m): 3

Enter the value of r2 (m): 6

"Capacitance of Isolated Sphere (F) = "
6.676D-10

"Capacitance of Concentric Spheres (F) = "
6.676D-10

-->
```

Theoretical calculation:

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Determination and plotting of Capacitance of Isolated and concentric Sphere with different inner and outer radius and dielectric constant of the dielectric medium.

Output:

Enter the r1 value is: 3 mm

Enter the r2 value is: 6 mm

Practical output:

"Capacitance of Isolated Sphere"

$C = 6.676 \times 10^{-10}$

"Capacitance of Concentric Spheres"

$C = 6.676 \times 10^{-10}$

Theoretical calculation:

$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$

$C = 4\pi\epsilon_0 r$

$C = 4\pi (8.854 \times 10^{-12}) (3 \times 10^{-3})$

$C = 3.3379 \times 10^{-12} \text{ F}$

$C = 0.3338 \text{ pF}$

Capacitance of an isolated sphere

$C = 0.3338 \text{ pF}$

Capacitance of Concentric Spheres

$C = 4\pi\epsilon_0 \frac{r_1 r_2}{r_2 - r_1}$

$r_1 = 3 \times 10^{-3} \text{ m}$

$r_2 = 6 \times 10^{-3} \text{ m}$

$C = 4\pi (8.854 \times 10^{-12}) \frac{(3 \times 10^{-3})(6 \times 10^{-3})}{(6 \times 10^{-3}) - (3 \times 10^{-3})}$

$C = 4\pi (8.854 \times 10^{-12}) (2 \times 10^{-3})$

$C = 6.6758 \times 10^{-12} \text{ F}$

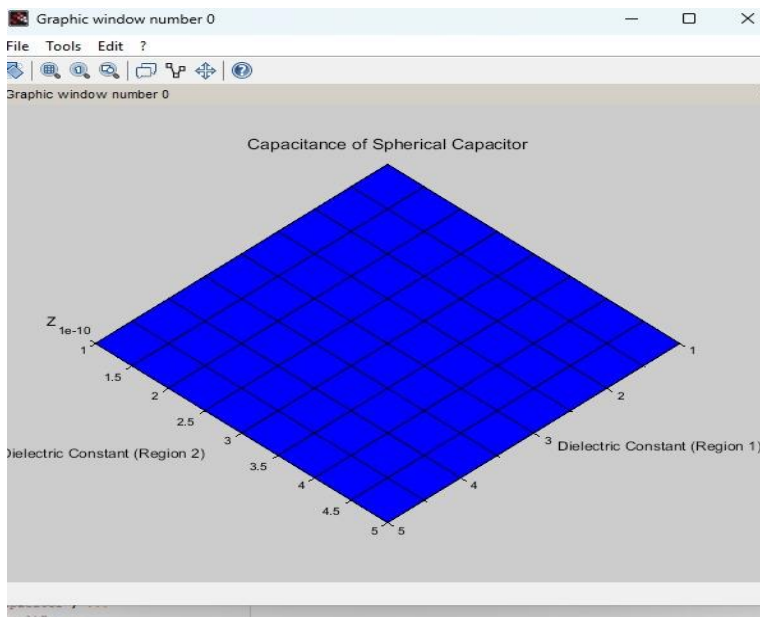
$C = 0.6676 \text{ pF}$

CASE 1:-CONCENTRIC SPHERICAL CAPACITOR WITH DIFFERENT DIELECTRIC MEDIUMS

Program:

```
EX 7.scf (C:\Users\user\EX 7.scf) - SciNotes
File Edit Format Options Window Execute ?
EX 7.scf (C:\Users\user\EX 7.scf) - SciNotes
EX 7.scf
epsilon_0 = 8.854e-12;
// Radii (m)
R1 = 0.2;
R2 = 0.6;
R3 = 1.0;
// Relative permittivity values
epsilon_r1 = 1.0;
epsilon_r2 = 1.0;
// Create mesh
[X,Y] = meshgrid(epsilon_r1, epsilon_r2);
// Initialize capacitance matrix
C = zeros(X);
// Calculate capacitance
for i = 1:size(X,1)
    for j = 1:size(X,2)
        x1 = X(i,j);
        x2 = Y(i,j);
        C1 = 4 * pi * epsilon_0 * x1 * (R1 * R2) / (R2 - R1);
        C2 = 4 * pi * epsilon_0 * x2 * (R2 * R3) / (R3 - R2);
        // Equivalent capacitance
        C(i,j) = (C1 * C2) / (C1 + C2);
    end
end
// 3-D Surface Plot
plot3D(epsilon_r1, epsilon_r2, C);
title("Capacitance of Spherical Capacitor", ...
    "Dielectric-Constant (Region-1)", ...
    "Dielectric-Constant (Region-2)");
axis grid;
```

Output:

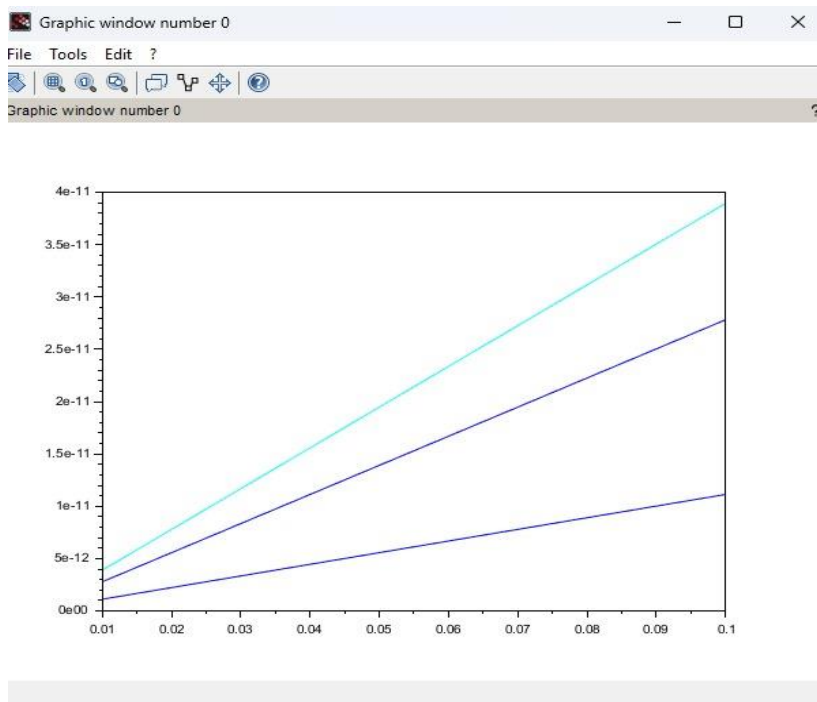


CASE 2: ISOLATED SPHERE

Program:

```
EX 7.sci (C:\Users\user\EX 7.sci) - SciNotes
File Edit Format Options Window Execute ?
EX 7.sci (C:\Users\user\EX 7.sci) - SciNotes
EX 7.sci
2 clear;
3 clf;
4
5 // Radius from 0.01 m to 0.1 m
6 R = linspace(0.01, 0.1, 100);
7
8 // Relative permittivity values
9 epsilon = [1 2.5 3.5];
10
11 // Permittivity of free space
12 epsilon0 = 8.854e-12;
13
14 // Initialize capacitance matrix
15 C = zeros(3, length(R));
16
17 // Calculate capacitance
18 for i = 1:3
19     C(i,:) = 4 * pi * epsilon0 * epsilon(i) .* R;
20 end
21
22 // Plot graphs
23 plot(R, C(1,:));
24 plot(R, C(2,:));
25 plot(R, C(3,:));
26
27 // Change line colors
28 e = gca();
29 e.children(1).foreground = 4; // Blue
30 e.children(2).foreground = 2; // Green
31 e.children(3).foreground = 3; // Red
32
33 // Labels
34 xlabel("Radius - (m)");
35 ylabel("Capacitance - (F)");
36 title("Capacitance of Isolated Sphere vs Radius");
37
38 // Legend
39 legend(["ε = 1 (Air)", "ε = 2.5", "ε = 3.5"]);
40
41 // Grid
42 grid();
```

Output:

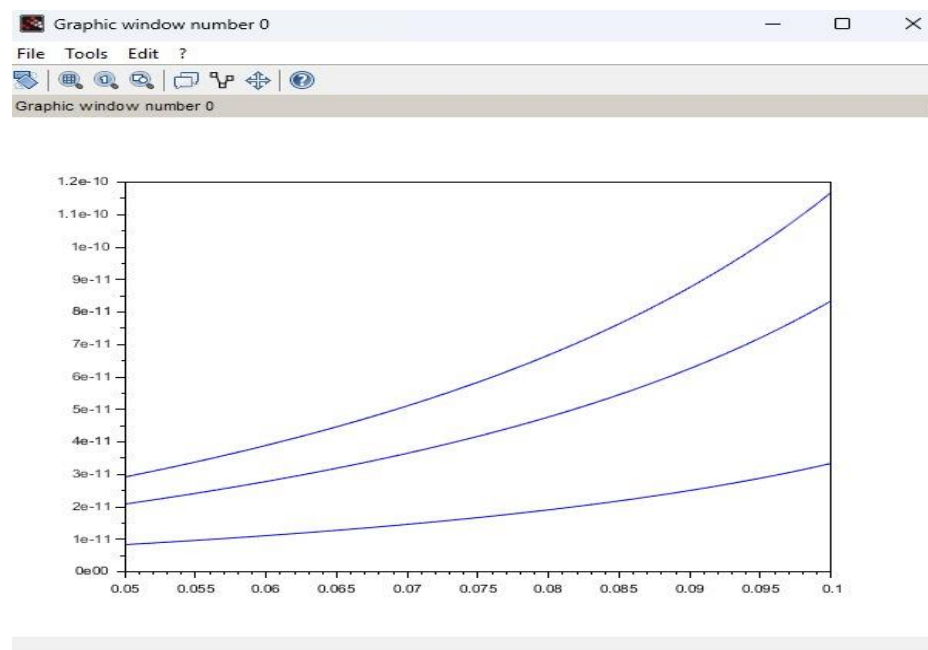


CASE 3: CONCENTRIC SPHERICAL CAPACITOR

Program:

```
EX 7.sci (C:\Users\user\EX 7.sci) - SciNotes
File Edit Format Options Window Execute ?
EX 7.sci (C:\Users\user\EX 7.sci) - SciNotes
EX 7.sci
8 //Inner-radius-(m)
9 R1 = linspace(0.05, 0.10, 100);
10
11 //Outer-radius-(constant)
12 R2 = 0.15;
13
14 //Relative-permittivity-values
15 epsilon = [1 2.5 3.5];
16
17 //Initialise-capacitance-matrix
18 C = zeros(3, length(R1));
19
20 //Calculate-capacitance
21 for i = 1:3
22     C(i,:) = (4 * %pi * epsilon0 * epsilon(i) * R1 * R2) ./ (R2 - R1);
23 end
24
25 //Plot-first-curve
26 plot(R1, C(1,:));
27
28 //Plot-remaining-curves
29 plot(R1, C(2,:));
30 plot(R1, C(3,:));
31
32 //Set-different-colors
33 a = gca();
34 a.children(1).foreground = 4; // Blue
35 a.children(2).foreground = 3; // Green
36 a.children(3).foreground = 2; // Red
37
38 //Labels
39 xlabel("Inner-Radius- (m)");
40 ylabel("Capacitance- (F)");
41 title("Capacitance-of-Concentric-Spherical-Capacitor-vs-Inner-Radius");
42
43 //Legend
44 legend(["Air-(x=-1)", "x=-2.5", "x=-3.5"]);
45
46 //Grid
47 grid();
48
```

Output:



Result:

Thus the program was verified for Capacitance of an isolated sphere and two concentric sphere using SCILAB was successfully.